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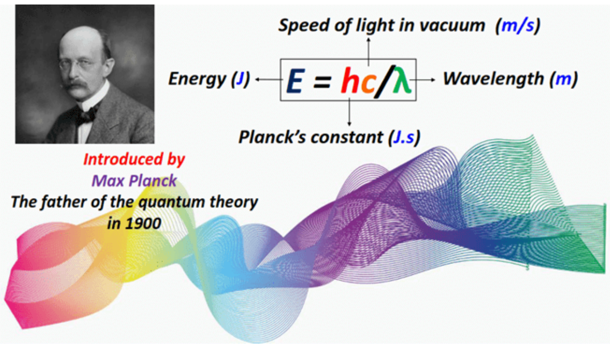
E= hc/wavelength in chemistry, what is it, where to use it?



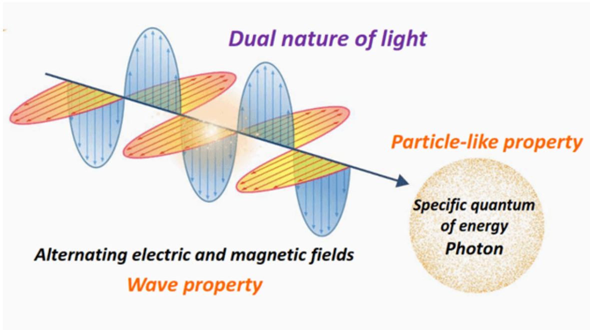
E = hc/λ is a fundamental equation that provides insight into the ***dual nature of light*** by highlighting its ***energy-wavelength relationship.***

Light is a form of electromagnetic radiation, visible to the human eye.

E = hc/λ explains that light behaves both as a wave as well as a particle.

Light propagates as a wave, possessing electric and magnetic fields, oscillating perpendicular to each other.

At the same time, light is composed of a large number of photons containing discrete packets of energy called quanta.



In the equation E = hc/λ, lambda (λ) or wavelength justifies the wave-like characteristics of light. In contrast, energy (E) denotes the particulate nature of light.

In this article, we will teach you how to use E= hc/λ also written as E= hc/wavelength or E= hc/lambda to solve plenty of numerical questions.

But before that, let's get introduced to the components of E= hc/λ one by one.

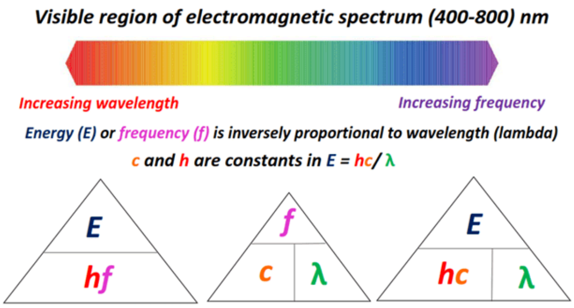
Components of E=hc/λ

In the equation **E= hc/λ or E=hc/lambda:**

- **E** = energy of a photon. Photon is a particle representing a quantum of light.
- **h** = Planck's constant. The value of Planck's constant stays fixed, i.e., **6.63×10^{-34} J.s.**
- **c** = speed of light, i.e., **3.0×10^{8} m/s**.
- **λ (lambda)** = wavelength of light.

As the values of c and h stay constants in E=hc/ λ, this implies that the energy of a photon is inversely proportional to the wavelength of light and vice versa.

Visible light ranges from a wavelength of **400-800 nm.** The greater the wavelength, the lower its frequency which in turn means that its photons possess lower energy individually.



Units of E, h, c, and λ in E=hc/λ

In E=hc/λ or E=hc/wavelength;

- The **energy (E)** is measured in **Joules (J)**
- **Speed of light (c)** is measured in **meters per second (m/s)**
- **Wavelength (λ) or lambda** is given in **meters (m)**

Therefore, the unit of Planck's constant (h) is calculated as follows:

Make h the subject of the formula from E=hc/λ

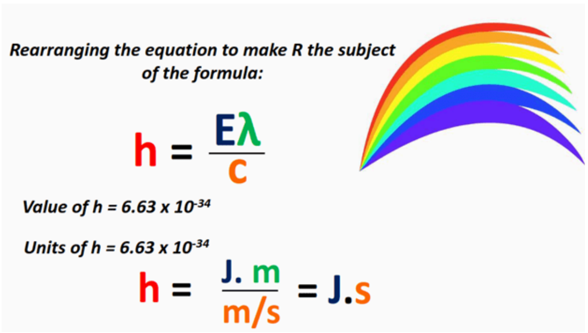
$$h = \frac{E\lambda}{c}$$

Substitute the units of E, λ and c in the above expression:

$$h = \frac{(J)(m)}{(\frac{m}{s})}$$

$$h = \frac{(J)(\cancel{m})(s)}{(\cancel{m})}$$

∴ The units of **h** are **Joule.sec (J.s)**



Where to use E=hc/wavelength – Examples

As h and c are constant values in E=hc/λ so we can use this formula to determine either the unknown wavelength or energy of light, depending on which variable is unknown.

For example, In the Balmer series of hydrogen, an electronic transition occurs from the third energy level (n=3) to the second energy level (n=2). This electronic transition is accompanied by an emission of visible light of wavelength 656.3 nm.

Calculate the energy emitted via the transition.

Solution

E= hc/λ explains the energy-wavelength relationship.

The wavelength (λ) is given in the question statement, **λ = 656.3 nm.**

Do not forget to convert nanometer (nm) into meter (m) to keep consistency in units, as shown below:

1 m = 10^9 nm

1 nm = 10^{-9} m

656.3 nm = 656.3 x 10^{-9} = **6.563 x 10^{-7} m**

We know the constant values already, i.e., **h= 6.63 x 10^{-34} J.s and c = 3.0 x 10^8 m/s.**

Hence, now we can find the energy emitted by substituting the above variables in:

$$\Rightarrow E = \frac{hc}{\lambda}$$

$$E = \frac{(6.63 \times 10^{-34})(3.0 \times 10^{8})}{(6.563 \times 10^{-7})}$$

∴ E = 3.03 x 10^{-19} J.

Result: The energy of the light emitted by the electron undergoing a transition from n= 3 to n=2 equals **3.03×10^{-19} J.**

*Now let us see another example where we will teach you how to find wavelength if the energy is given using **E =hc/λ.***

Another example is- A photon has an energy of 4.65×10^{-19} J; find its corresponding wavelength.

Solution

In this example, the wavelength (λ) is unknown while the corresponding energy (E) is given in the question statement, i.e., **$E = 4.65 \times 10^{-19}$ J.**

Therefore, the energy-wavelength relationship formula E=hc/λ can be rearranged to make wavelength the subject, as follows:

$$\Rightarrow \lambda = \frac{hc}{E}$$

Now let us substitute the known variable along with the constants, i.e., h= 6.63×10^{-34} J.s and c = 3.0×10^{8} m/s, into the above expression.

$$\lambda = \frac{(6.63 \times 10^{-34})(3.0 \times 10^{8})}{(4.65 \times 10^{-19})}$$

$$\therefore \lambda = 4.28 \times 10^{-7} \text{ m.}$$

Result: The wavelength of light is **4.28×10^{-7} m** or **428 nm.**

Let's try another example- How many kilojoules per mole (kJ/mol) of energy is contained in a photon of wavelength 550 nm?

Solution

As per the question statement:

λ (wavelength) = 550 nm = **5.5×10^{-7} m**

We already know:

h= **6.63×10^{-34} J.s**

c = **3.0×10^{8} m/s**

Now let's plug in all the above data in the equation:

$$E = \frac{hc}{wavelength}$$

$$E = \frac{(6.63 \times 10^{-34})(3.0 \times 10^{8})}{(5.5 \times 10^{-7})}$$

$$\therefore E = 3.62 \times 10^{-19} \text{ J.}$$

Do not forget that we need to convert Joules into Kilojoules, as shown below:

1 J = 10^{-3} kJ

E = 3.62×10^{-19} = $3.62 \times 10^{-19} \times 10^{-3}$ = **3.62×10^{-22} kJ.**

As a final step, we need to convert the above answer into kJ/mol.

∴ **1 mole of photon = Avogadro number of particles = 6.022×10^{23}**

Therefore, we need to multiply the energy obtained in kJ with the Avogadro number:

E = (3.62×10^{-22}) (6.022×10^{23})

∴ **E = 218.0 kJ/mol**

Result: The energy of the photon of wavelength 550 nm is **218.0 kJ/mol.**

FAQ

What is E=hc/lambda?

E=hc/lambda explains the dual nature of light. It denotes the inverse relationship between the wavelength of light and the energy possessed by its photons.

In E=hc/λ:

- **E** = **Energy** of a photon
- **h** = **Planck's constant** (6.63×10^{-34} J.s)
- **c** = **speed of light** (3.0×10^{8} m/s)
- **λ (lambda)** = **Wavelength** of light

What is Planck's constant (h) in E=hc/wavelength?

Planck's constant (h) is a fundamental constant in quantum mechanisms that relates the energy of a photon with the frequency or wavelength of light.

The value of Planck's constant stays fixed in E =hc/wavelength, i.e., **h = 6.63×10^{-34} J.s.**

Due to a change in wavelength, does the energy of light change while traveling from one medium to another medium?

As per E=hc/wavelength, the energy (E) is **inversely proportional** to wavelength (λ).

Thus, changing wavelength **does change** the energy. The lower the wavelength of light, the higher its energy, and vice versa.

How is the frequency (f) related to energy (E) as per E=hc/λ?

Frequency (f) is inversely related to the wavelength (λ) of light, as shown below:

$$f = c/\lambda$$

Replacing c/λ with f in E=hc/λ gives us **Planck's equation**:

$$E = hf$$

As per the above equation, the energy possessed by its photons is **directly proportional** to the frequency of a light wave.

Did you like it?

SUGGEST CORRECTIONS IF ANY 0

About the author

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Ammara Waheed is a highly qualified and experienced chemist, whose passion for Chemistry is evident in her writing. With a Bachelor of Science (Hons.) and Master of Philosophy (M. Phil) in Physical and Analytical Chemistry from Government College University (GCU) Lahore, Pakistan, with a hands-on laboratory experience in the Pakistan Council of Scientific and Industrial Research (PCSIR), Ammara has a solid educational foundation in her field. She comes from a distinguished research background and she documents her research endeavors for reputable journals such as Wiley and Elsevier. Her deep knowledge and expertise in the field of Chemistry make her a trusted and reliable authority in her profession. Let's connect - https://www.researchgate.net/profile/Ammara-Waheed

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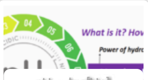
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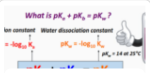
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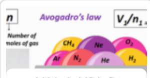
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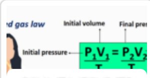
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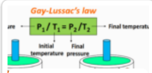
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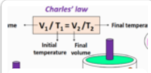
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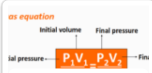
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